TIME INTELLIGENCE ANALYSIS

Bicycle Sales Data — Power BI Portfolio Project

Comprehensive time-series analysis using DAX Time Intelligence measures including YTD, YoY, MoM, Rolling 12-Month, Moving Averages & Seasonality

- Microsoft Power BI
- DAX
- Time Intelligence
- Calendar Table
- Data Modelling

Developed by Dr. Abraham Eninla

77.55M
Gross Revenue

1.31%
Profit Margin

35.13M
YoY Growth

0.83
Sales Growth %

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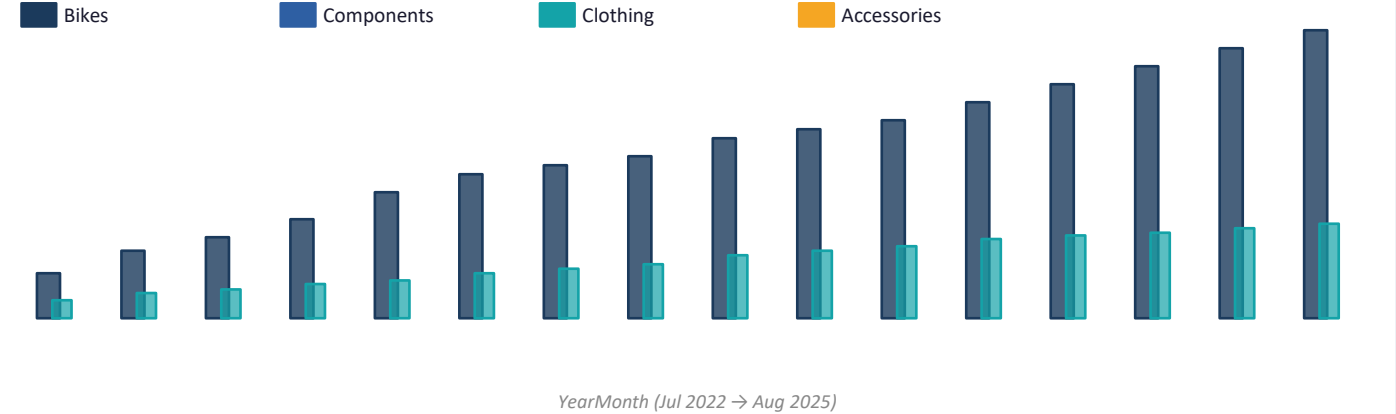
Sales Trend by Category & Time Intelligence KPI Dashboard

Full Period: July 2022 – August 2025

77.55M	**76.55M**	**999.49K**	**0.01**	**-240.63K**
Gross Revenue	Cost of Goods Sold	Net Profit	Net Profit Margin	Profit Growth
SUM Total Sales	*Total Cost*	*Total Profit*	*Profit Margin %*	*vs Previous Year*
77.55M	**35.13M**	**12.65M**	**35.13M**	**0.83**
MoM Growth	YoY Growth	Salesperson YTD	Sales Growth	Sales Growth %
Month-on-Month	*Year-on-Year*	*Year-to-Date*	*Absolute Growth*	*Growth Rate*

Sales Trend by Category — Line Chart (YearMonth Axis | Bikes, Components, Clothing, Accessories)



📊 KEY INSIGHTS

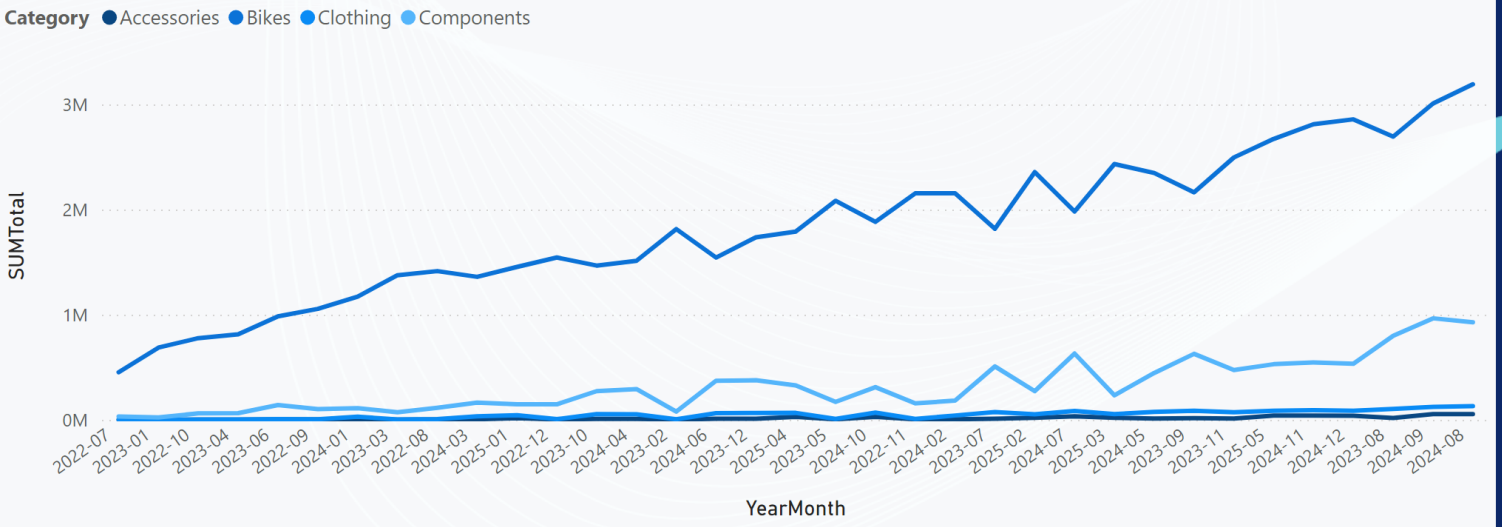
- Bikes dominate with sustained upward trajectory across all 3 years
- YoY Growth of 35.13M confirms strong revenue expansion
- Profit Growth of -240.63K signals cost escalation outpacing revenue
- MoM Growth = 77.55M shows momentum in latest period
- Profit Margin at 0.01 (1%) — critically thin requiring urgent action
- Salesperson YTD of 12.65M shows strong individual contributions

Source: Power BI Time Intelligence Dashboard | LinkedIn Portfolio Analysis

77.55M	76.55M	999.49K	0.01	-240.63K
SUMTotal	Total Cost	Total Profit	Profit Margin	Profit Growth

77.55M	35.13M	12.65M	35.13M	0.83
MoM Growth	YoY Growth	Salesperson YTD	Sales Growth	Sales Growth %

Sales Trend by Category



Filter By Quarter, Year

Year, Quarter: All

- 2022
- 2023
- 2024
- 2025

Filter By Category

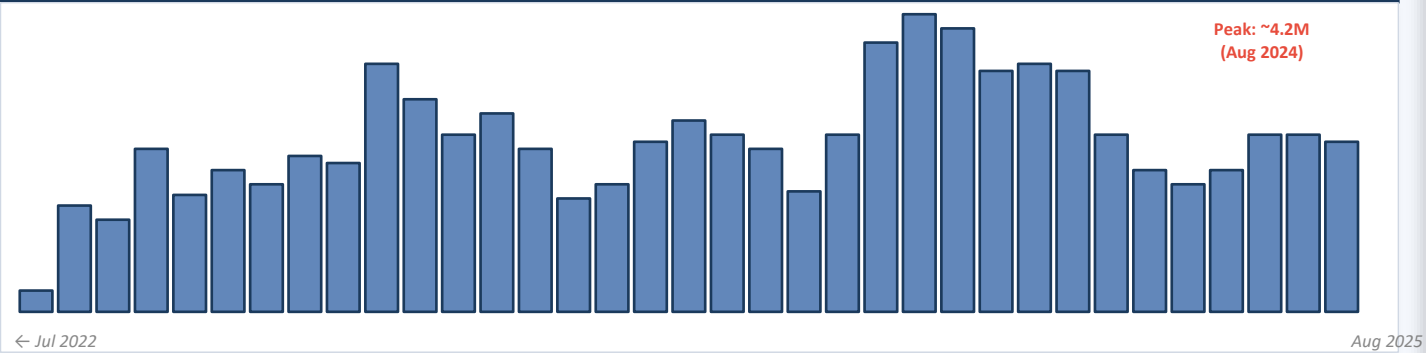
Category: All

- Accessories
- Bikes
- Clothing
- Components

Total Sales & Total Profit Monthly Trend Analysis

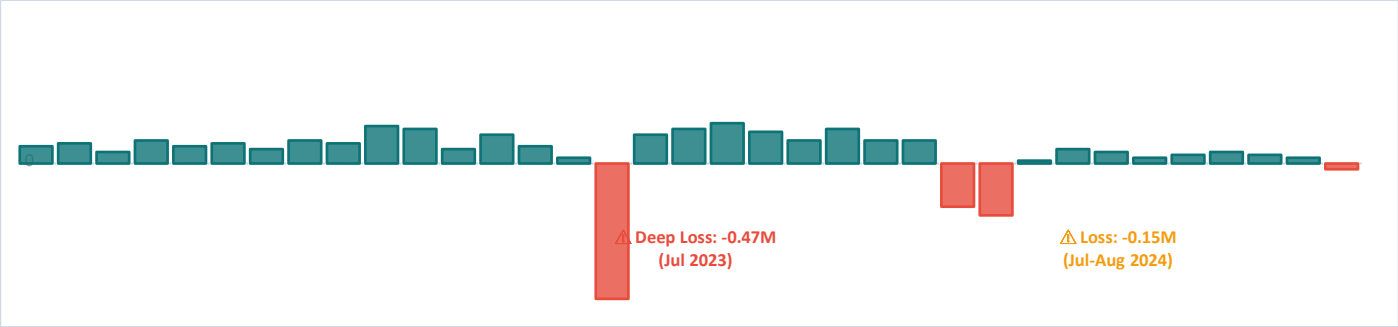
Jul 2022 – May 2025

Total Sales (Gross Revenue) by Year-Month



Y-Axis: SUM Total ($M) | X-Axis: YearMonth

Total Profit (Net Income) by Year-Month — Including Negative Profit Periods



Y-Axis: Total Profit ($M) | Positive = Green | Negative = Red | X-Axis: YearMonth

ANALYSIS

- Sales climbed from ~$0.3M (Jul 2022) to a peak of ~$4.2M (Aug 2024) — a 14x growth trajectory.
- Two significant profit dips identified: July 2023 (-$0.47M) and July–August 2024 (-$0.15M to -$0.18M), requiring cost investigation.
- The 2025 period shows modest sales of $2–2.5M with thin profits — post-peak stabilisation phase.
- High sales volatility with sharp month-to-month swings suggests sporadic bulk orders rather than stable demand.

DECISIONS

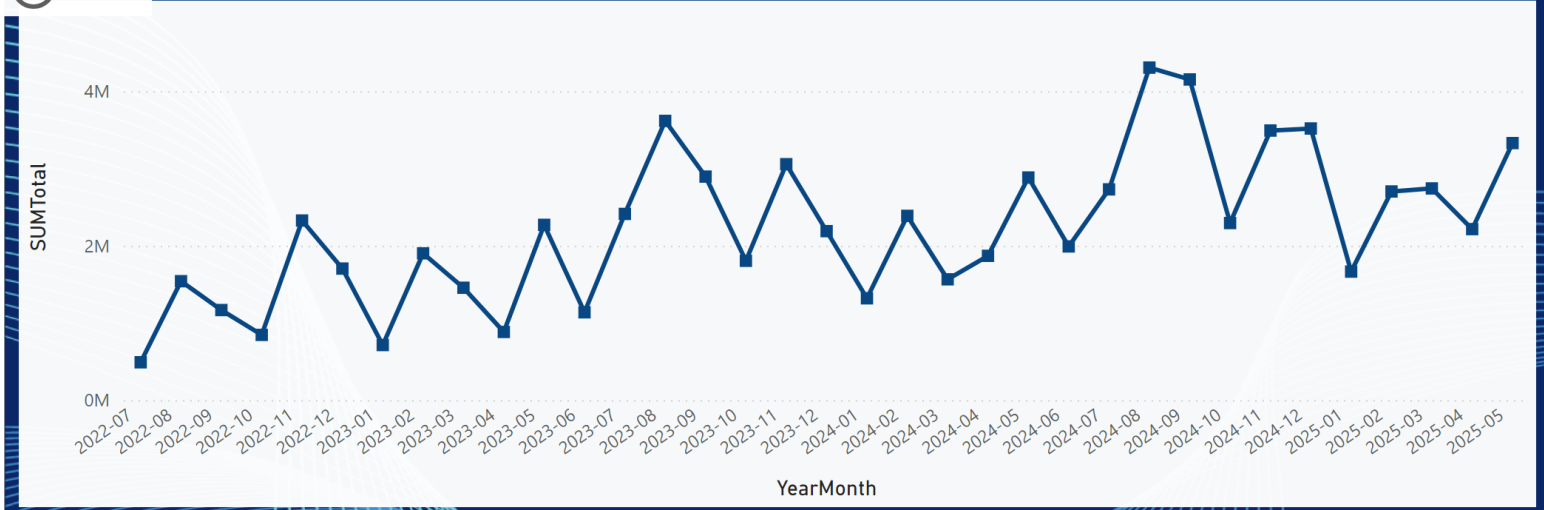
1. Investigate July 2023 cost spike — identify the root cause of the -$0.47M profit crash and prevent recurrence.

2. Establish monthly profit floor KPI: set -$50K as an automatic alert threshold in the dashboard.

3. Analyse peak months (Aug 2024) to identify the drivers and replicate those conditions.

Source: Power BI Time Intelligence Dashboard

Total Sales by YearMonth

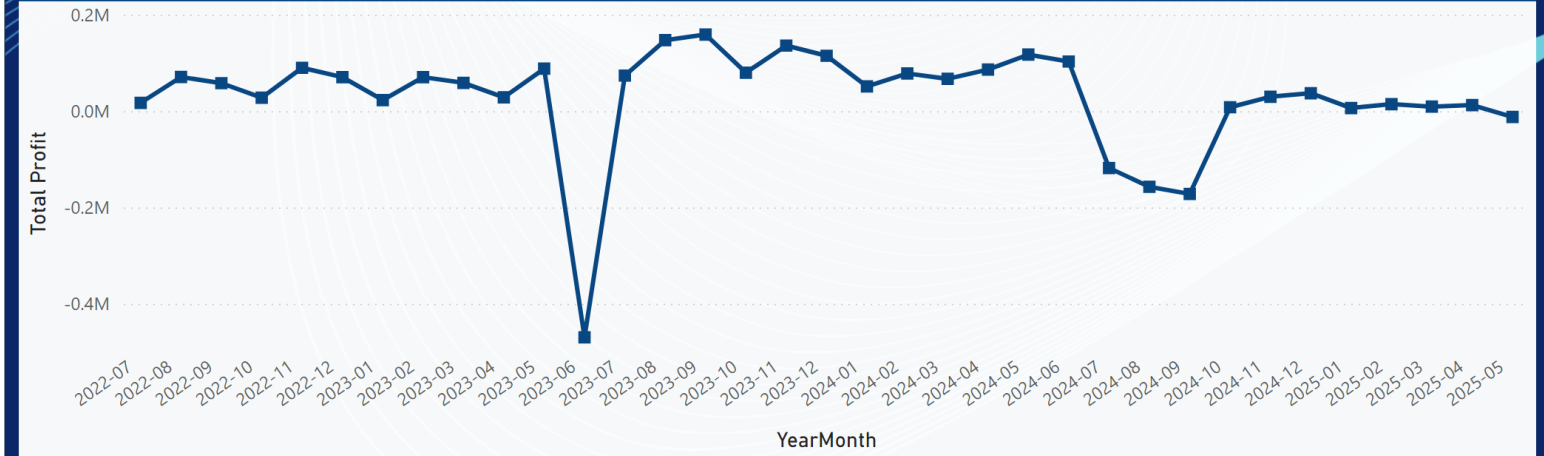


Filter Report

Month

- Select all
- April
- August
- December
- February
- January
- July
- June
- March
- May
- November
- October
- September

Total Profit by YearMonth



Slice Report by Year

Year: All

- 2022
- 2023
- 2024
- 2025

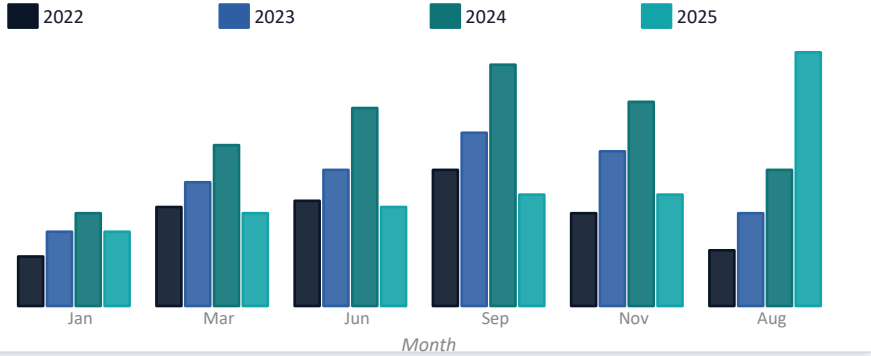
Slice Report by

Quarter: All

- Q1
- Q2
- Q3
- Q4

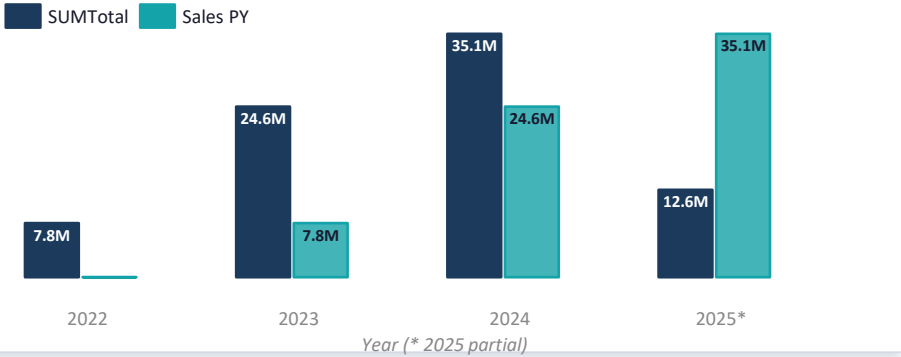
Multi-Year Comparison, Rolling 12-Month Trend & Sales Seasonality

Total Sales by Month & Year (Clustered Column)



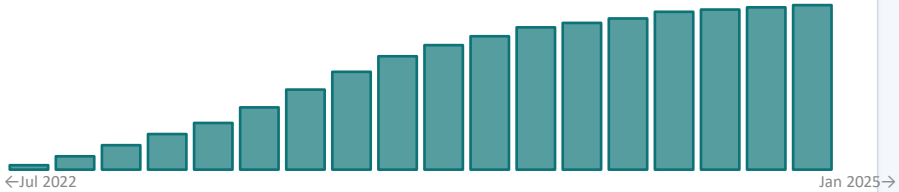
Key Insight: 2024 shows the strongest monthly performance across nearly every month, with August peaking at ~$4.1M. The 2025 partial year is already trending ahead of 2022 and 2023 for the same months, confirming upward revenue trajectory.

Year-on-Year Comparison: Current Sales vs Previous Year Sales



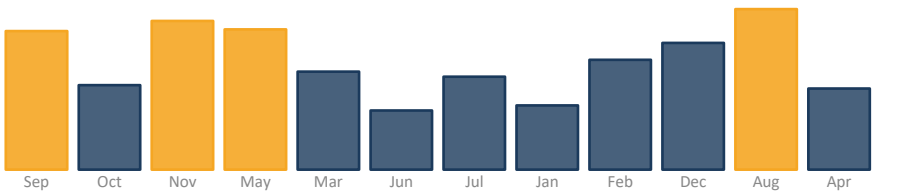
Key Insight: Revenue grew from $7.8M (2022) to $35.1M (2024) — a 350% growth in 2 years. YoY Growth measure of 35.13M confirms 2024 as the breakthrough year. 2025 partial data at $12.6M is pacing ahead of prior years for the same period.

Rolling 12-Month Sales by Date — Smoothed Revenue Trend



Rolling 12M Sales ($M) | Date Axis — confirms sustained long-term growth with no reversals

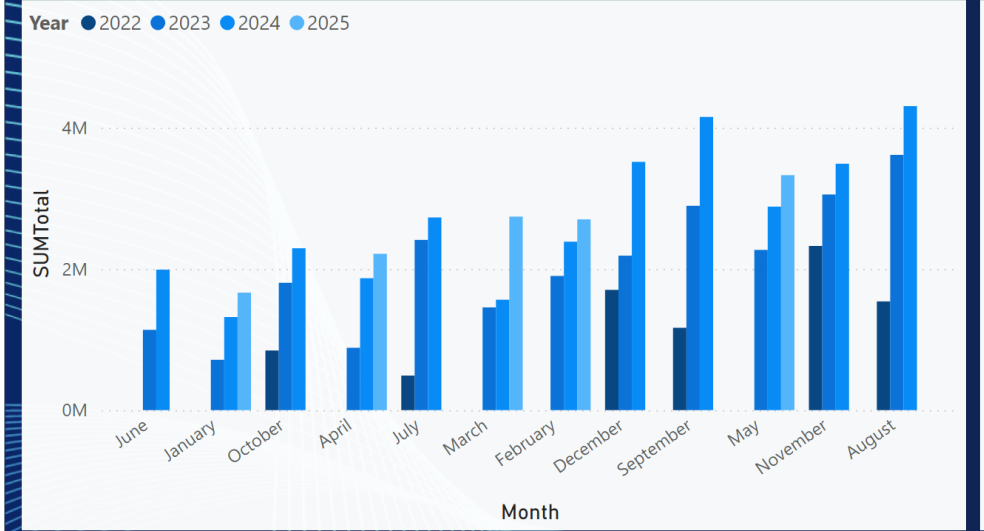
Seasonality Chart — Total Sales by Month (Pattern Across All Years)



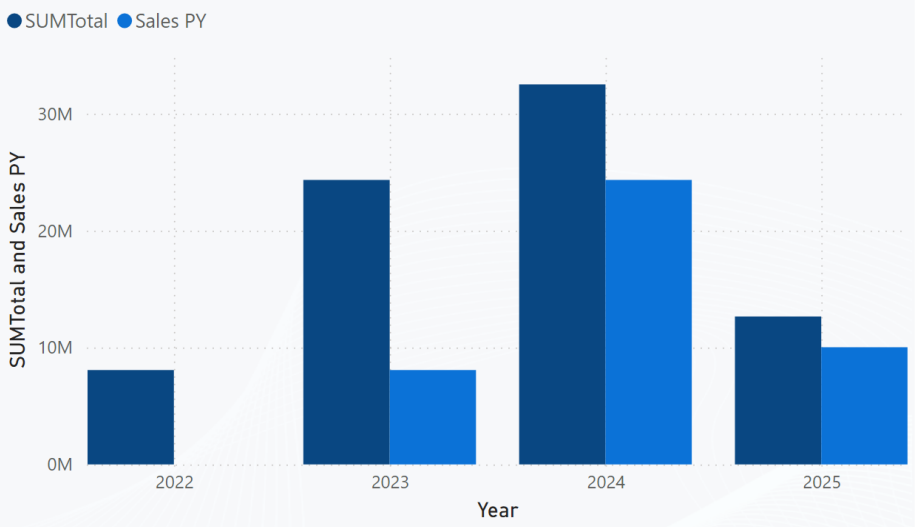
Highlighted (Gold) = Peak Sales Months | Month Axis

Source: Power BI Time Intelligence Dashboard | LinkedIn Portfolio Analysis

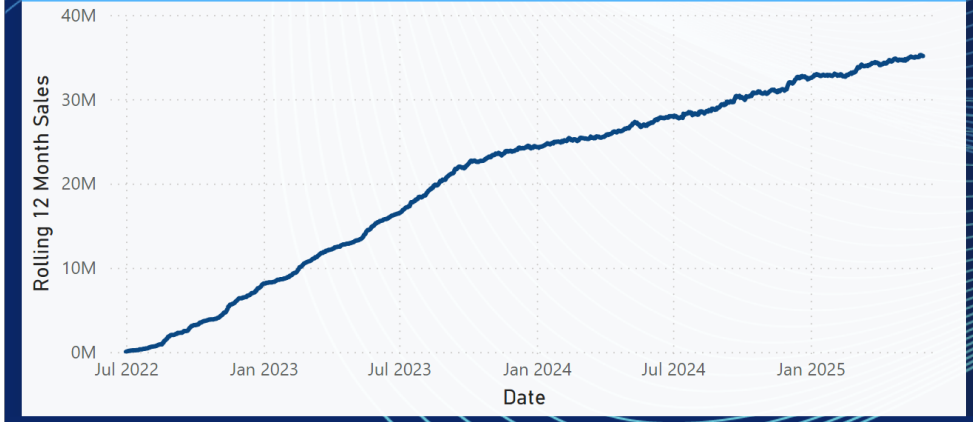
Total Sales by Month and Year



Year on Year Comparison



Rolling 12 Month Sales by Date



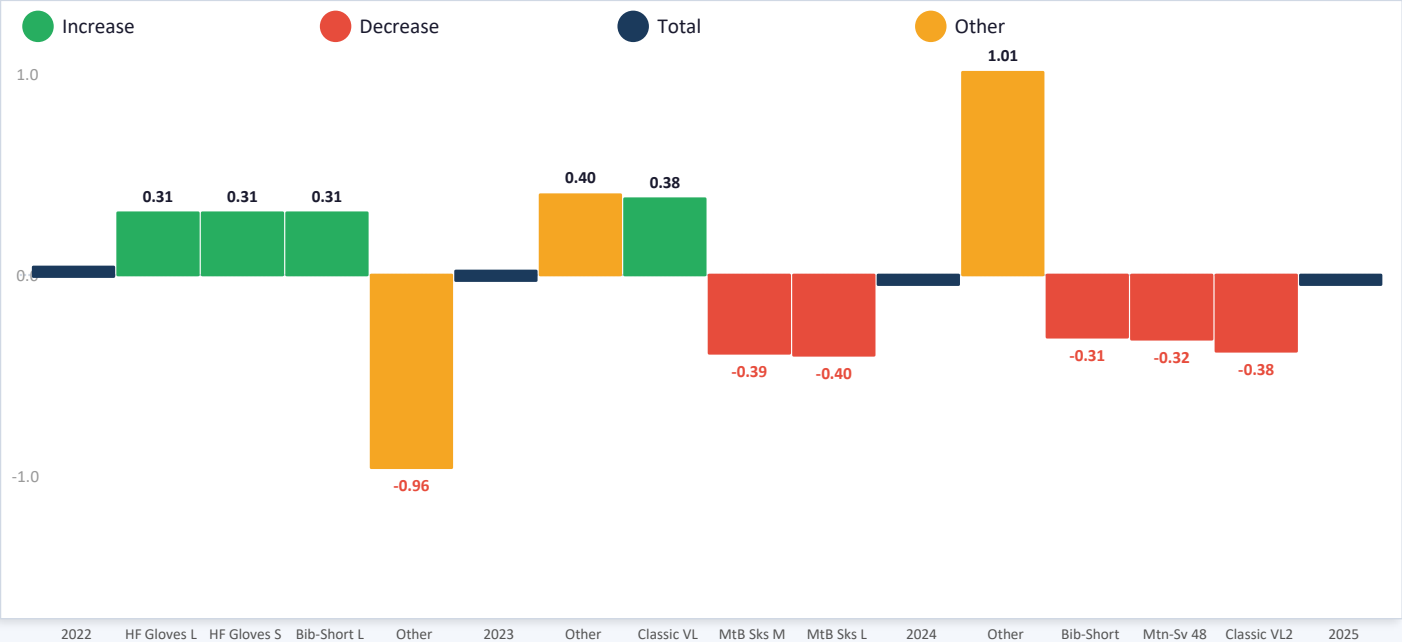
Seasonality

Waterfall Chart: Profit Margin Analysis by Product & Year

Measure Slicer:

MoM Growth | **Profit Margin** | SUMTotal | Total Cost | Total Profit | Total Quantity | YoY Growth | Sales Growth | Rolling 12M | 3M Moving Avg

Parameter Slicer: **Product** | Category | Country | Region



Profit Margin | X-Axis: Year / Product | Green=Increase Red=Decrease Gold=Other Navy=Total

Source: Power BI Waterfall Chart | Measure = Profit Margin | Slicer = Product

ANALYSIS

- 2022 total margin was just 0.04 (4%), driven upward by Half-Finger Gloves and Bib-Shorts, then severely eroded by the 'Other' group (-0.96).
- 2023 shows margin near zero (0.02), with Other products adding 0.40 but Mountain Bike Socks both sizes pulling margins negative (-0.39 and -0.40).
- 2024 margin collapsed to 0.00 despite 'Other' products contributing +1.01 — completely offset by three products each destroying -0.31 to -0.38.
- The waterfall reveals specific product-level margin destroyers, providing actionable intelligence for pricing or discontinuation decisions.

DECISIONS

1. Immediately review Mountain Bike Socks pricing — both variants are consistent margin destroyers across 2023 and 2024.

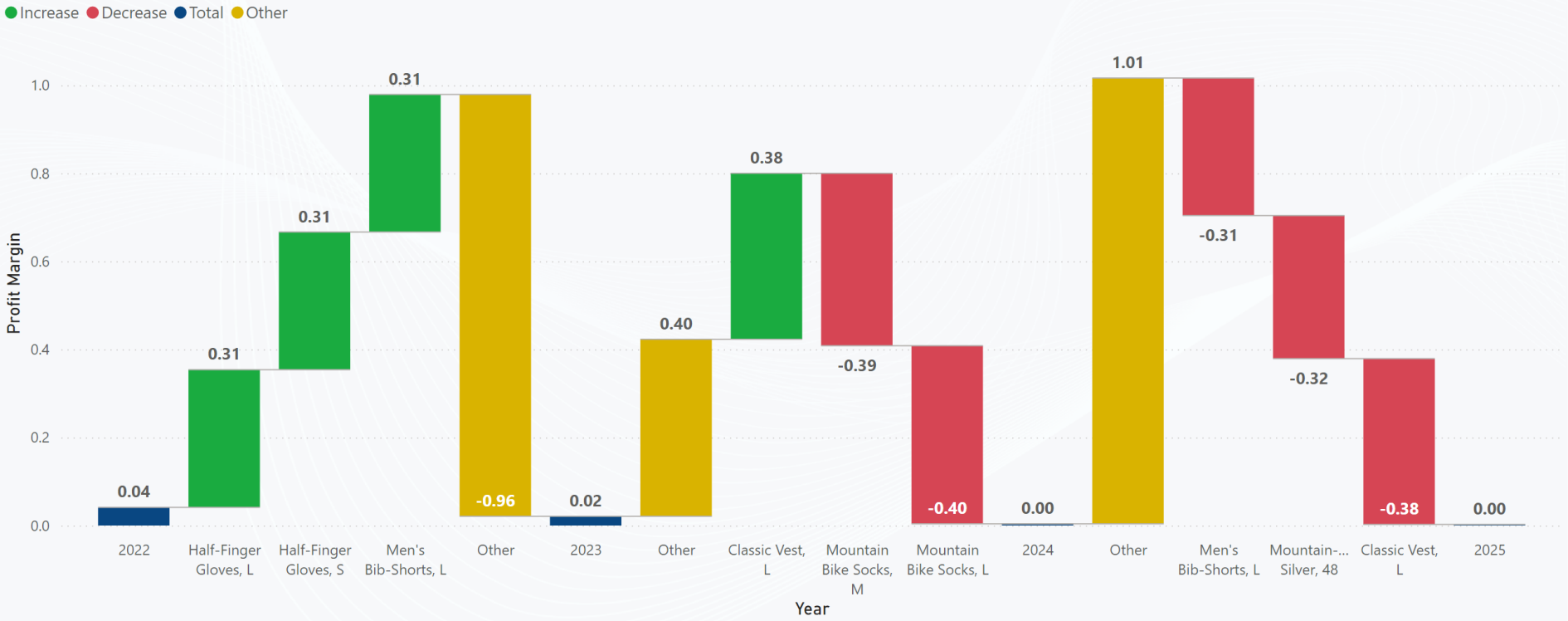
2. Review and reprice Classic Vest L — appears as a drag in 2024. Evaluate production cost vs market price.

Measure for Waterfall Chart

MoM Growth	Profit Margin	SUMTotal	Total Cost	Total Profit	Total Quantity	YoY Growth	Sales Growth	Rolling 12 Month Sales	3 Month Moving Avg

Product	Category	Country	Region

'Sales'[Profit Margin]



SLIDE 6 — Data Quality Assessment: Column Distribution, Profile & Quality

Salesperson

Rows: 18 | Cols: 5

Valid 100% | Error 0% | Empty 0%

18 distinct EmployeeKeys (272–296). All columns 100% valid. Perfect dimension table quality.

Date Table

Rows: 1,461 | Cols: 4

Valid 100% | Error 0% | Empty 0%

12 distinct months, 3 years, MonthNum avg 6.1. Well-structured calendar. No gaps detected.

Products

Rows: 397 | Cols: 10

Valid 100% | Error 0% | Empty 0%

397 unique ProductKeys. Standard Cost min $0.86 max $2,171.29, avg $436.82. 4 categories, 37 subcategories.

Region

Rows: 10 | Cols: 4

Valid 100% | Error 0% | Empty 0%

10 territories: 5 US regions + Canada, France, Germany, Australia, UK. 3 groups: North America, Europe, Pacific.

Sales (Qty)

Rows: 3,616 | Cols: 10

Valid 100% | Error 0% | Empty 0%

Quantity: min 1, max 21, avg 2.51. Right-skewed distribution — most orders are 1–3 units.

Sales (Unit Price)

Rows: 3,616 | Cols: 10

Valid 100% | Error 0% | Empty 0%

Unit Price: min $5.01, max $2,146.96, avg $724.79. Wide range confirms multi-tier product pricing.

Sales (Cost)

Rows: 3,616 | Cols: 10

Valid 100% | Error 0% | Empty 0%

Cost: min $3.40, max $26,770.16, avg $1,713.64. High std dev ($2,693.88) reflects mixed low/high-cost SKUs.

🏆 OVERALL DATA QUALITY VERDICT

All 5 tables passed 100% column quality validation with zero errors and zero empty values. The dataset is production-ready with no imputation or cleaning required. The star-schema structure is correctly implemented with proper primary keys (EmployeeKey, ProductKey, SalesTerritoryKey, Date) enabling high-performance DAX calculations and reliable Time Intelligence measures.

Source: Power Query — Column Distribution, Column Quality & Column Profile Analysis

DASHBOARD KPI NAMING & DAX MEASURES REFERENCE

Professional analytics titles for your dashboard + All Time Intelligence measures at a glance

Total Sales

Gross Revenue

Industry-standard P&L term for top-line sales before deductions. Signals financial literacy to stakeholders and recruiters.

Total Profit

Net Profit / Net Income

Universally recognised bottom-line metric. 'Net Income' is preferred in formal financial reporting contexts.

Total Cost

Cost of Goods Sold (COGS)

Standard P&L expense line. Positions the KPI within a formal income statement framework, essential for executive stakeholders.

Time Intelligence DAX Measures — Complete Reference

Sales YTD *YTD cumulative sales tracking*	**Sales PY** *Previous year same period*	**YoY Growth** *Absolute year-on-year growth*
YoY Growth % *Percentage growth vs prior year*	**Sales PM** *Prior month sales*	**MoM Growth** *Month-over-month absolute growth*
MoM Growth % *Month-over-month % change*	**Rolling 12M Sales** *Trailing 12-month revenue*	**3M Moving Avg** *3-month smoothed average*
Salesperson YTD *Sales rep YTD performance*	**Profit YTD** *Cumulative YTD profit*	**Best Month** *Identifies highest sales month*